About This Book

Who should read this book

Deep Reinforcement Learning in Action is a course designed to take you from the very foundational concepts in reinforcement learning all the way to implementing the latest algorithms. As a course, each chapter centers around one major project meant to illustrate the topic or concept of that chapter. We've designed each project to be something that can be efficiently run on a modern laptop; we don't expect you to have access to expensive GPUs or cloud computing resources (though access to these resources does make things run faster).

This book is for individuals with a programming background, in particular, a working knowledge of Python, and for people who have at least a basic understanding of neural networks (a.k.a. deep learning). By "basic understanding," we mean that you have at least tried implementing a simple neural network in Python even if you didn't fully understand what was going on under the hood. Although this book is focused on using neural networks for the purposes of reinforcement learning, you will also probably learn a lot of new things about deep learning in general that can be applied to other problems outside of reinforcement learning, so you do not need to be an expert at deep learning before jumping into deep reinforcement learning.

How this book is organized: A roadmap

The book has two sections with 11 chapters.

Part 1 explains the fundamentals of deep reinforcement learning.

- Chapter 1 gives a high-level introduction to deep learning, reinforcement learning, and the marriage of the two into deep reinforcement learning.